Path spaces of an n-plus-one truncation are n-truncated path spaces

The Univalent Foundations Program

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Editorial note. Only the selected result is counted. Its local constructions and author proof are retained without generated proof completion.

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Allowed standard prerequisites (editorial).

- Univalent homotopy theory foundations. Univalence, function extensionality, dependent path transport, higher inductive pushouts/suspensions, truncations and basic pointed homotopy notions. The selected local code, fiber or gluing constructions are retained as author proof. Original numbered references identify these foundations.

Source reference. Original section 2.12: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1625–1625. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 2.13: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1626–1626. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 167–167. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original theorem 7.1.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1793–1793. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 2.3.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1894–1894. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 2.3.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1750–1750. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

We can characterize the path spaces of a truncation using the same method that we used in original sections 2.12 and 2.13 for coproducts and natural numbers (and which we will use in original section 8 to calculate homotopy groups). Unsurprisingly, the path spaces in the $(n+1)$-truncation of A are the n-truncations of the path spaces of A. Indeed, for any $x, y : A$ there is a canonical map

$$f : \left\|x =_A y\right\|_n \to \left(|x|_{n+1} =_{\|A\|_{n+1}} |y|_{n+1}\right) \tag{0.0.1}$$

defined by

$$f(|p|_n) :\equiv \mathsf{ap}_{|-|_{n+1}}(p).$$

This definition uses the recursion principle for $\|-\|_n$, which is correct because $\|A\|_{n+1}$ is $(n+1)$-truncated, so that the codomain of f is n-truncated.

Theorem 0.0.2. *For any A and $x, y : A$ and $n \geq -2$, the map* (0.0.1) *is an equivalence; thus we have*

$$\left\|x =_A y\right\|_n \simeq \left(|x|_{n+1} =_{\|A\|_{n+1}} |y|_{n+1}\right).$$

Proof. The proof is a simple application of the encode-decode method: As in previous situations, we cannot directly define a quasi-inverse to the map (0.0.1) because there is no way to induct on an equality between $|x|_{n+1}$ and $|y|_{n+1}$. Thus, instead we generalize its type, in order to have general elements of the type $\|A\|_{n+1}$ instead of $|x|_{n+1}$ and $|y|_{n+1}$. Define $P : \|A\|_{n+1} \to \|A\|_{n+1} \to n\text{-}\mathsf{Type}$ by

$$P(|x|_{n+1}, |y|_{n+1}) :\equiv \|x =_A y\|_n$$

This definition is correct because $\|x =_A y\|_n$ is n-truncated, and n-Type is $(n+1)$-truncated by original theorem 7.1.11. Now for every $u, v : \|A\|_{n+1}$, there is a map

$$\mathsf{decode} : P(u, v) \to \left(u =_{\|A\|_{n+1}} v\right)$$

defined for $u = |x|_{n+1}$ and $v = |y|_{n+1}$ and $p : x = y$ by

$$\mathsf{decode}(|p|_n) :\equiv \mathsf{ap}_{|-|_{n+1}}(p).$$

Since the codomain of decode is n-truncated, it suffices to define it only for u and v of this form, and then it's just the same definition as before. We also define a function

$$r : \prod_{u:\|A\|_{n+1}} P(u, u)$$

by induction on u, where $r(|x|_{n+1}) :\equiv |\mathsf{refl}_x|_n$.

Now we can define an inverse map

$$\mathsf{encode} : (u =_{\|A\|_{n+1}} v) \to P(u, v)$$

by

$$\mathsf{encode}(p) :\equiv \mathsf{transport}^{v\mapsto P(u,v)}(p, r(u)).$$

To show that the composite

$$(u =_{\|A\|_{n+1}} v) \xrightarrow{\mathsf{encode}} P(u,v) \xrightarrow{\mathsf{decode}} (u =_{\|A\|_{n+1}} v)$$

is the identity function, by path induction it suffices to check it for $\mathsf{refl}_u : u = u$, in which case what we need to know is that $\mathsf{decode}(r(u)) = \mathsf{refl}_u$. But since this is an $(n-1)$-type, hence also an $(n+1)$-type, we may assume $u \equiv |x|_{n+1}$, in which case it follows by definition of r and decode. Finally, to show that

$$P(u,v) \xrightarrow{\mathsf{decode}} (u =_{\|A\|_{n+1}} v) \xrightarrow{\mathsf{encode}} P(u,v)$$

is the identity function, since this goal is again an $(n-1)$-type, we may assume that $u = |x|_{n+1}$ and $v = |y|_{n+1}$ and that we are considering $|p|_n : P(|x|_{n+1}, |y|_{n+1})$ for some $p : x = y$. Then we have

$$\begin{aligned}
\mathsf{encode}(\mathsf{decode}(|p|_n)) &= \mathsf{encode}(\mathsf{ap}_{|-|_{n+1}}(p)) \\
&= \mathsf{transport}^{v\mapsto P(|x|_{n+1},v)}(\mathsf{ap}_{|-|_{n+1}}(p), |\mathsf{refl}_x|_n) \\
&= \mathsf{transport}^{y\mapsto \|x=y\|_n}(p, |\mathsf{refl}_x|_n) \\
&= \left|\mathsf{transport}^{y\mapsto (x=y)}(p, \mathsf{refl}_x)\right|_n \\
&= |p|_n,
\end{aligned}$$

using original lemmas 2.3.10 and 2.3.11. (Alternatively, we could do path induction on p; the desired equality would then hold judgmentally.) This completes the proof that decode and encode are quasi-inverses. The stated result is then the special case where $u = |x|_{n+1}$ and $v = |y|_{n+1}$. □